PIONEER ACADEMICS

RESEARCH PROJECT

Analysis of Periodicity in COVID-19 Daily New Cases Using Discrete Fourier Transform

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PIONEER ACADEMICS

Abstract

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by Jiahao PENG

The purpose of this study was to perform frequency domain analysis on COVID-19 data, and potentially aid the government in planning public health interventions. COVID-19 time series were transformed into the frequency domain and plotted on periodograms for data visualization. In this study, weekly, annual, and biannual periodicities were found, and this paper provides interpretations and explanations of each periodicity. This paper also presents a detailed explanation and derivation of Discrete Fourier Transform Formula and Fast Fourier Transform Algorithm.

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List of Abbreviations

DFT	**D**iscrete **F**ourier **T**ransform
WHO	**W**orld **H**ealth **O**rganization
FFT	**F**ast **F**ourier **T**ransform
COVID-19	**Co**rona**vi**rus **D**isease of 20**19**
IDFT	**I**nverse **D**iscrete **F**ourier **T**ransform

List of Symbols

x_t	input signal amplitude at time t
X_n	spectrum of signal at frequency n
N	number of time samples = number of frequency samples
i	imaginary number $\sqrt{-1}$
e	Euler's number $\lim_{n\to\infty}(1+\frac{1}{n})^n = 2.718281828459045...$
A_n	amplitude of signal at frequency n
θ_n	phase angle of signal at frequency n

Chapter 1

Introduction

1.1 Background

The ongoing global pandemic of COVID-19 first began on December 2019, and the World Health Organization (WHO) declared COVID-19 as a pandemic in March 2020 (Rahimi et al., 2021). By 2021, many variants of the virus have appeared, and some have become dominant in many countries. There have been more than 220 million confirmed cases and 4.6 million confirmed deaths as of September 2020. The pandemic has resulted in severe social and economic disruptions, including the largest global recession since the Great Depression in the 1930s.

1.2 Literature Review

Yoshiyasu Takefuji, analyzing COVID-19 daily deaths in the United States, demonstrated a 7-day cycle of the mortality data (Takefuji, 2021). The author suggested that the weekly periodicity reflects variation in US weekly reporting activity related to human and societal factors, rather than true variation in daily deaths. No true epidemic patterns, such as seasonality, have been verified in Yoshiyasu Takefuji's study, mainly due to the lack of epidemic data available at the time of publish, Mar 4, 2021. Since Yoshiyasu Takefuji was only able to gather data on 238 days, from March 1, 2019 to October 24, 2020, he was not able to reveal the epidemic patterns of COVID-19: for seasonal variations to manifest themselves clearly in the frequency domain, the sample size needs to be at least one year. However, with the extended data size of 627 days currently available online, the seasonality analysis that was previously implausible is now performable.

1.3 Periodicity of Epidemics

Within the broad area of epidemiologic inquiry, a small but important body of research has focused on a theory of seasonality, that is, the premise that outbreaks and epidemics correlate to recurrent weather patterns. It has been shown that disease outbreaks occur more during winter months; the reason behind this seasonality remains unclear, with suggestions ranging from temperature influences to seasonal changes in host physiology(Eckrode, 2010).

Seasonality, whatever the cause, can be viewed as a component of another larger phenomenon, that is, periodicity, the frequency-dependent recurrence of epidemics. Of note, different geographic areas might be impacted by seasonality to various extents due to varying climates. Consequently, in examining periodicity of the COVID-19 pandemic, seasonality of data in different geographic areas might exhibit differing strengths.

It is unclear whether the eradication of COVID-19, zero incidence of infection globally, can be achieved, so understanding pandemic patterns is essential in preventing large outbreaks in the coming years. Currently, the overall basic seasonal trend of COVID-19 is understood, but policymakers still lack quantitative information and analysis needed to forecast the future trend of COVID-19. The government proposes policy responses of different severity as the infectiousness of the disease fluctuates periodically; therefore, information on disease epidemic characteristics over time can help guide prevention strategies, such as vaccinating in anticipation of an outbreak.

More radical response measures, such as large-scale quarantine and economic shutdown, cause greater costs to the economy relative to more conservative measures, such as mask and social-distancing mandates. Consequently, duration of radical preventive measures such as lockdown should be minimized through analysis and planning, and costly measures should only be imposed at an optimal time prior to an outbreak for maximum efficacy. Minimizing the duration and maximizing the efficacy of economic shutdown reduces both the number of infected and the negative side effects of the lockdown impact, including low GDP growth and deterioration of citizens' mental health.

1.4 Problem Statement and Objectives

The research aims to use a quantified way to elucidate patterns of the COVID-19 pandemic; specifically, using time-series analysis to measure the periodic recurrences of COVID-19 outbreaks. The results can aid public health administrators in longterm planning by providing predictive models. When a new unknown coronavirus variant emerges, such virus adaptations may be unacknowledged; however, the new variant and the epidemic's deviation from the normal pattern can be quickly identified through comparing the interpolated daily new cases and the actual data. If an epidemic pattern changes, or if a new variant emerges, this change in disease dynamics can be quickly spotted through frequency domain analysis, and analysts can immediately alert practitioners to a change in epidemiology of COVID.

1.5 Summary of Results

Weekly, annual, and biannual periodicities were found in this study, and vaccination is confirmed to be powerful and beneficial in limiting the spread of COVID-19.

Chapter 2

Method and Theoretical Base

The Fourier Transform transforms signals from time domain to frequency domain; in other words, it decomposes signals into oscillatory components. The Inverse Fourier Transform transforms signals from frequency domain to time domain; simply put, it rebuilds the signals from oscillatory components. In the frequency domain, the phase and amplitude of the component sinusoids can be directly examined. Since the data for COVID-19 is time-discrete, the Discrete Fourier Transform will be used.

The Discrete Fourier Transform can be calculated with two nested loops running through all data points in a program, thus the runtime of the program is $O(N^2)$. Fortunately, using the Fast Fourier Transform algorithm, the time required to calculate a DFT can be reduced to $O(N \log N)$.

2.1 Discrete Fourier Transform

2.1.1 Bases in $\mathbb{C}^N$

Suppose the data points of COVID-19 data are given as

$$(0, x_0), (1, x_1), ..., (N-1, x_{N-1})$$

The goal is to represent the set of data in terms of orthogonal oscillatory basis vectors in $\mathbb{C}^N$, in other words, transform data from time domain to frequency domain. This transformation, which will be obtained and explained in detail in Discrete Fourier Transform, is done through using a different set of orthogonal bases to represent the data vector. Viewing from the time domain, the N data points can be represented through nonoscillatory orthogonal bases

$$E_0 = (1, 0, ..., 0)\ E_1 = (0, 1, ..., 0)\ .....\ E_{N-1} = (0, 0, ..., 1)$$

Then, every x_t (t representing time passed since the start of epidemic) in the dataset can be represented by

$$x_t = \sum_{n=0}^{N-1} X_n \cdot E_n \quad \Big(t = 0, 1, ..., N-1\Big)$$

Evidently, with the chosen bases E_n, X_n is simply equal to x_t, because the calculation of x can also be viewed as the matrix multiplication of X with an identity matrix.

Above is the representation of data points with nonoscillatory bases, but the goal, transforming the data to frequency domain, requires representation of data points with oscillatory bases.

First of all, define inner product in $\mathbb{C}^N$

$$\langle f, g \rangle = \sum_{n=0}^{N-1} f_n \cdot \overline{g_n}$$

Define a set of basis vectors (which will be later proved to be orthogonal and periodic)

$$S = \{e_0, e_1, ..., e_{N-1}\}, \quad e_n^t = e^{2\pi i \frac{nt}{N}} \quad (t = 0, 1, ..., N-1)$$

The variable t represents the time, and n is the index of frequency.

2.1.2 Proof of Orthogonality

Definition 1. *A set of vectors $\{\phi_0, \phi_1, ..., \phi_{N-1}\}$ is an orthogonal set of vectors if any two vectors in the set are orthogonal to each other, so*

$$\langle \phi_n, \phi_m \rangle \ = \ 0 \quad (m \neq n)$$

Lemma 1. *$\{e_n\}_{n=0}^{N-1}$ is an orthogonal basis in $\mathbb{C}^N$*

Proof.

$$\begin{aligned}
\langle \phi_n, \phi_m \rangle &= \sum_{t=0}^{N-1} (e^{2\pi i \frac{nt}{N}}) \overline{(e^{2\pi i \frac{mt}{N}})} \\
&= \sum_{t=0}^{N-1} e^{2\pi i \frac{n-m}{N} t} \\
&= \sum_{t=0}^{N-1} \left[e^{2\pi i \frac{n-m}{N}} \right]^t
\end{aligned}$$

When $n - m = 0$:

$$\langle \phi_n, \phi_m \rangle_{m=n} = \sum_{t=0}^{N-1} (e^0)^t = \sum_{t=0}^{N-1} 1 = N$$

When $n - m \neq 0$: notice the series resulting from the inner product forms a geometric progression with common ratio $r = e^{2\pi i \frac{n-m}{N}}$. The geometric series can be calculated:

$$\begin{aligned}
\langle \phi_n, \phi_m \rangle_{m \neq n} &= \frac{1 - (e^{2\pi i (n-m)})}{1 - e^{2\pi i \frac{n-m}{N}}} \quad \text{Note: } e^{2\pi i} = 1 \\
&= \frac{1 - (1^{n-m})}{1 - e^{2\pi i \frac{n-m}{N}}} \\
&= \frac{0}{1 - e^{2\pi i \frac{n-m}{N}}} \\
&= 0
\end{aligned}$$

In conclusion:

$$\langle \phi_n, \phi_m \rangle = \begin{cases} N & m = n \\ 0 & m \neq n \end{cases}$$

□

2.1.3 Proof of Periodicity

Lemma 2. *e_n is N-periodic* $\equiv e_n^t = e_n^{t+N}$

Proof.

$$\begin{aligned} e_n^{t+N} &= e^{\frac{2\pi i n}{N}(t+N)} \\ &= e^{2\pi i \frac{nt}{N}} \cdot e^{2\pi i n} \quad \text{Note: } e^{2\pi i} = 1 \\ &= e^{2\pi i \frac{nt}{N}} \\ e_n^{t+N} &= e_n^t \end{aligned}$$

□

2.1.4 Representation of Points by Orthogonal Vectors

Suppose x_t is a vector of data points to be represented by the linear combination of an arbitrary orthogonal set of vectors $\{\phi_n\}$. Because $\mathbb{C}^N$ is an N-dimensional vector space and $\{\phi_n\}$ is an orthogonal set with N elements, the set $\{\phi_n\}$ spans the whole vector space $\mathbb{C}^N$, thus, any vector in $\mathbb{C}^N$ is representable through the linear combination of the orthogonal set of basis vectors $\{\phi_n\}$.

$$x_t = \sum_{n=0}^{N-1} X_n \phi_n \tag{2.1}$$

Then, the scalars X_n can be easily determined due to the linearly independent property of the elements in $\{\phi_n\}$.

Lemma 3.

$$X_n = \frac{\langle x, \phi_n \rangle}{\langle \phi_n, \phi_n \rangle} \tag{2.2}$$

Proof. Pick some ϕ_n, and find inner product with x:

$$\begin{aligned} \langle x, \phi_n \rangle &= \sum_{t=0}^{N-1} x_t \cdot \overline{\phi_n^t} \\ &= \sum_{t=0}^{N-1} \Big(\sum_{n=0}^{N-1} X_m \phi_m^t \Big) \overline{\phi_n^t} \quad \text{interchange order of summation} \\ &= \sum_{m=0}^{N-1} X_m \sum_{t=0}^{N-1} \phi_m^t \overline{\phi_n^t} \\ &= \sum_{m=0}^{N-1} X_m \langle \phi_m, \phi_n \rangle \quad \text{Note: } \langle \phi_n, \phi_m \rangle_{m \neq n} = 0 \\ \langle x, \phi_n \rangle &= \langle \phi_n, \phi_n \rangle \cdot X_n \\ X_n &= \frac{\langle x, \phi_n \rangle}{\langle \phi_n, \phi_n \rangle} = \frac{\langle x, \phi_n \rangle}{N} \end{aligned}$$

□

2.1.5 Discrete Fourier Transform Formulas

In the subsections Proof of Orthogonality and Proof of Periodicity, it s demonstrated that the set of vectors $\{e_n\}$, where $e_n^t = e^{2\pi i \frac{nt}{N}}$, is both orthogonal and periodic.

Representation of Points by Orthogonal Vectors obtains a generalized formula for the coefficients of basis vectors. Substituting the set of basis vectors $\{\phi_n\}$ with $\{e_n\}$ into equations (2.2) and (2.1) gives the equations for Discrete Fourier Transform and Inverse Discrete Fourier Transform.

$$DFT(x) = X_n = \frac{1}{N}\sum_{t=0}^{N-1} x_t \cdot e^{-2\pi i \frac{nt}{N}} \quad n = 0, 1, 2, ..., N-1 \tag{DFT}$$

$$IDFT(X) = x_t = \sum_{n=0}^{N-1} X_n \cdot e^{2\pi i \frac{nt}{N}} \quad t = 0, 1, 2, ..., N-1 \tag{IDFT}$$

The Discrete Fourier Transform is a function transforming an N-dimensional vector of complex numbers $\{x_t\}_{t=0}^{N-1}$ in the time domain into another sequence of complex numbers $\{X_n\}_{n=0}^{N-1}$ in the frequency domain.

X_n is a complex number that encodes both the amplitude and phase information of the complex sinusoidal component $e^{-2\pi i \frac{nt}{N}}$ (Hong, 2020). The amplitude and phase of the signal at frequency n can be calculated:

$$A_n = \mid X_n \mid = \sqrt{\mathrm{Re}(X_n)^2 + \mathrm{Im}(X_n)^2}$$
$$\theta_n = \arg(X_n)$$

2.2 Fast Fourier Transform

With equation (DFT), the Discrete Fourier Transform of a signal can be found in $O(N^2)$ time: N oscillatory components, with the coefficients of oscillatory components each taking $O(N)$ to calculate. However, practical implementations of the Discrete Fourier Transform are usually based on the Cooley–Tukey FFT Algorithm, which performs the Discrete Fourier Transform in $O(N \log N)$.

2.2.1 Cooley–Tukey FFT Algorithm

The Cooley–Tukey FFT Algorithm is a divide-and-conquer algorithm; there are three steps to a divide-and-conquer algorithm: (1) break down a problem into two or more subproblems (2) solve the subproblems independently (3) combine solutions of subproblems to obtain solution of the original problem.

By the definition of DFT (with the prefactor $\frac{1}{N}$ omitted for the simplicity of demonstration):

$$\begin{aligned} DFT(x) = X_n &= \sum_{t=0}^{N-1} x_t \cdot e^{-2\pi i \frac{nt}{N}} \qquad \text{Assuming N is even:} \\ &= \sum_{t=0}^{\frac{N}{2}-1} x_{2t} \cdot e^{-2\pi i \frac{n(2t)}{N}} + \sum_{t=0}^{\frac{N}{2}-1} x_{2t+1} \cdot e^{-2\pi i \frac{n(2t+1)}{N}} \\ &= \sum_{t=0}^{\frac{N}{2}-1} x_{2t} \cdot e^{-2\pi i \frac{nt}{\frac{N}{2}}} + e^{-2\pi i \frac{n}{N}} \cdot \sum_{t=0}^{\frac{N}{2}-1} x_{2t+1} \cdot e^{-2\pi i \frac{n}{\frac{N}{2}}} \end{aligned}$$

The original DFT can be broken down into two $\frac{N}{2}-$ length DFTs, one with even-indexed elements and one with odd-indexed elements(Wörner, 2008). The FFT simply reuses the results of half-length DFTs to calculate the original DFT (Cameron, 2021). Due to this self-referentiality, the DFT can be recursively solved:

$$FFT(x) = FFT(x_{even}) + e^{-2\pi i \frac{n}{N}} \cdot FFT(x_{odd})$$

x_{even} represents a vector of size $\frac{N}{2}$ with all the even indexed numbers in x; x_{odd} represents a vector of size $\frac{N}{2}$ with all the odd indexed numbers in x. $FFT(x_{even})$ and $FFT(x_{odd})$ are half the size of $FFT(x)$; the dissymmetry in dimensions is resolved by periodically extending the smaller FFTs.

As long as the problem size is a power of 2, this recursive definition of DFT can be reapplied to decompose the two half-length DFTs into four quarter-length DFTs, then to eight eighth-length DFTs etc. The recursion calls continue until the base case is reached: when $N = 1$, the vector x can no longer by broken down into even-indexed and odd-indexed terms. In the base case, the DFT of x is simply equal to x itself:

$$DFT(x) = X_n = \sum_{t=0}^{N-1} x_t \cdot e^{-2\pi i \frac{nt}{N}} \quad \text{When vector } x \text{ has size one:}$$

$$X = x \cdot e^{-2\pi i \frac{0 \cdot 0}{1}}$$

$$DFT(x) = x$$

2.2.2 Time Complexity Analysis of FFT

The problem of size N will be recurrently divided into two sub-problems of size $\frac{N}{2}$, and combining the solutions takes linear time. Hence the recurrence relation:

$$T(N) = 2 \cdot T(\frac{N}{2}) + N$$

This equation can be illustrated through a recursion binary tree, where each node represents a sub-problem (Figure 2.1).

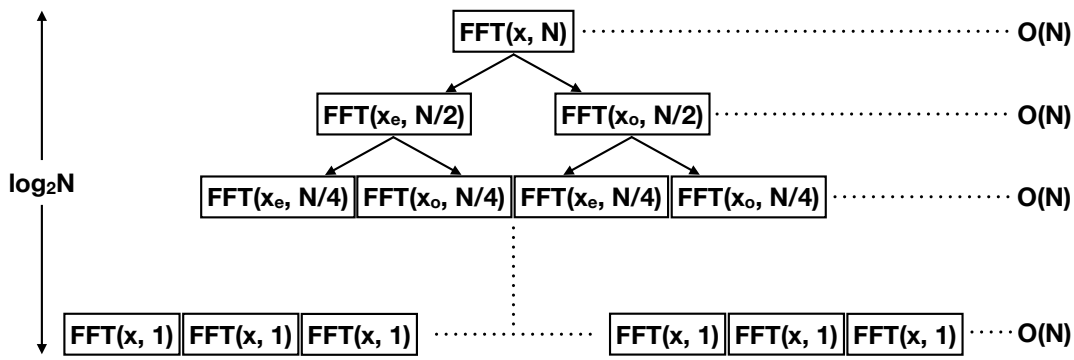

FIGURE 2.1: Recursion Tree of FFT.

Since layer n (root is layer 0) always consists of 2^n sub-problems, and each sub-problem has size $\frac{N}{2^n}$, the time cost of each level is always linear: $2^n \cdot \frac{N}{2^n} = N$. Suppose at level x, the size of sub-problems becomes 1, then

$$\frac{N}{2^x} = 1 \implies x = \log_2 N$$

$\therefore$ Total number of levels in the recursion tree is $\log_2 N + 1$

In conclusion, DFT calculation with FFT algorithm has time complexity $T(N) = O(N \cdot \log N)$.

2.2.3 Speed Comparison

To illustrate the enormous runtime improvement using the FFT algorithm, the running times for DFT and FFT functions, both implemented through python, are compared against each other with different input data sizes. It is also simply interesting to observe the efficiency of algorithms in practice. The input vector x was generated randomly in each test, and the timeit module was used to calculate the execution time, which was obtained by taking the average over ten runs. The running time will vary with different computers and compilers; the results in table 2.1 were acquired on a MacBook Air with a 1.2 GHz Quad-Core Intel Core i7 processor, and with a Python3 interpretor.

Input Size	$O(N^2)$ DFT Runtime	$O(N \cdot \log N)$ FFT Runtime
1	0.00001s	0.00001s
10	0.00003s	0.00001s
10^2	0.0002s	0.00001s
10^3	0.03s	0.00002s
10^4	4s	0.0001s
$2 \cdot 10^4$	40s	0.0002s
$3 \cdot 10^4$	160s	0.0004s
$4 \cdot 10^4$	400s	0.0005s
$5 \cdot 10^4$	750s	0.0007s
10^5	$>$ 1 hour (estimated)	0.002s
10^6	$>$ 1 week (estimated)	0.03s
10^7	$>$ 1 year (estimated)	0.6s
10^8	$>$ 100 years (estimated)	8s

TABLE 2.1: DFT and FFT Algorithm Runtime Comparison

Table 2.1 shows that both algorithms are efficient when the input size is small, but larger inputs bring out striking differences between the running times of algorithms with quadratic and linearithmic time complexity. For input sizes greater than 10^4, the Discrete Fourier Transform can be calculated instantly with FFT, whereas calculating without FFT would take an excessive amount of time.

Chapter 3

Investigation

Performing Fourier Transform on COVID-19 Daily New Cases data isolates the contribution of different frequency components to the pandemic, hence facilitating time-series analysis in frequency domain. By plotting the Fourier Transform results in a periodogram, periodicity of the pandemic can simply be observed by a novice.

Government policy changes, vaccine alterations, virus mutations, migration, replenishment of susceptible hosts, and changes in contact behavior throughout a year are all unplanned external factors expected to have notable impacts on the epidemic, bringing unstableness to the system (Eckrode, 2010). These external factors all manifest themselves in different ways in periodograms.

3.1 Data

The research uses WHO COVID-19 Global Data as the data set. This credible and authorized data set comprises of COVID-19 data officially reported by 237 countries and regions. Cases reflect confirmed cases, and dates represent dates of reporting.

It is possible for some countries to under-report their data,but it will not invalidate the possible periodicity of COVID-19; it will merely lower the amplitude of any detected periodic events. Thus, the discrepancy between reported cases and actual cases does not significantly impact the research results.

The data set contains daily counts of COVID-19 cases and deaths for 627 days, spanning January 3, 2020 to September 20, 2021, the date of publication.

Since the data set is a finite-length time-series, errors will occur at the beginning and end of the spectrum, as the Fourier Transform assumes the data is cyclic (Torrence and Compo, 1998). Additionally, this epidemiological time-series is short and noisy, therefore the impacts of external factors and boundary effects are enlarged(Cazelles et al., 2007). Ideally, the data set should encompass more than 5 years of data to reduce the relative effect of noises and better examine the overall trend; nevertheless, almost 2 years of data is small-scaled but sufficient to reveal some periodicity in COVID-19.

In the data set, COVID-19 information is organized into a Comma Separated Values (CSV) file. The possible relevent variables in the WHO COVID-19 Global Data are as follows:

- Date Reported (string) (coded as: year-month-day)
- Country (string) (region name)
- New Cases (integer) (new confirmed cases)
- New Deaths (integer) (new confirmed deaths)

3.2 Results

The purpose of this research is to analyze the periodicity of COVID-19 daily new cases data. The Discrete Fourier Transform of the data was calculated to enable frequency domain analysis of the presence of periodic patterns such as seasonality in COVID-19; periodicity is evidently observed through visualizing the frequency domain of COVID-19 data with periodograms. COVID data, stratified by country/region, was converted through numpy module's FFT function, and then plotted as periodograms using the matplotlib module.

3.2.1 Zero Day Period

The periodogram for COVID-19 daily new cases in the United States is shown in Figure 3.1.

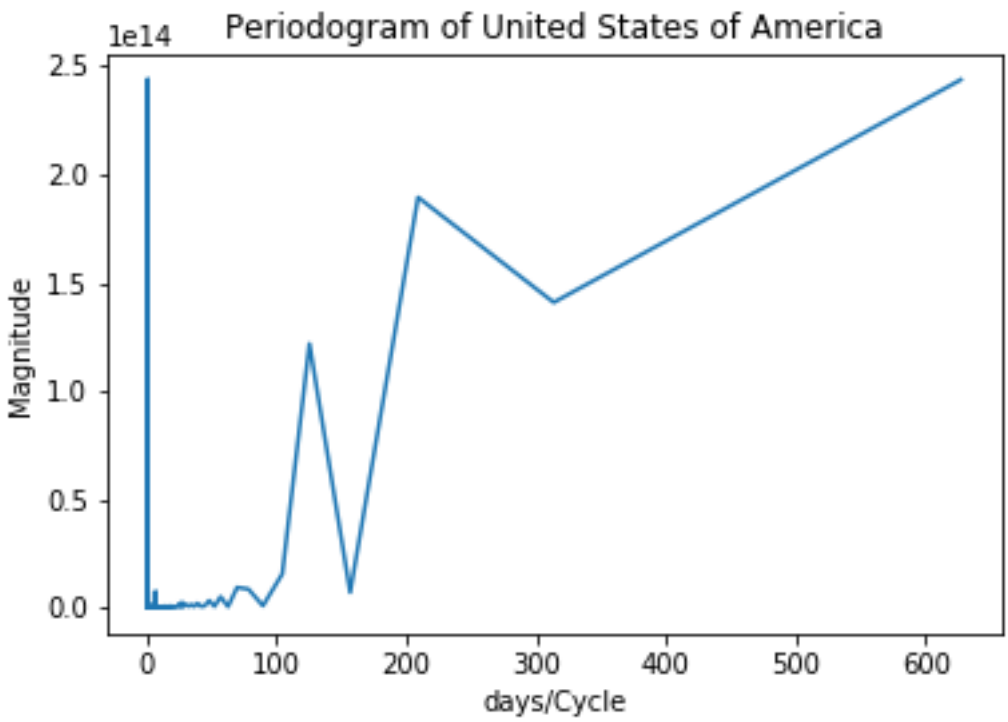

FIGURE 3.1: Periodogram of USA Daily New COVID-19 Cases

This periodogram exhibits a peak of approximate magnitude 10^{14} at period of 0 days per cycle; such "oddity" is also present in any other periodogram generated in this research. Recall the Discrete Fourier Transform Formula (DFT) (with prefactor $\frac{1}{N}$ omitted):

$$X_n = \sum_{t=0}^{N-1} x_t \cdot e^{-2\pi i \frac{nt}{N}}$$

$$\text{When } n = 0: \quad X_0 = \sum_{t=0}^{N-1} x_t$$

Therefore, the peak at 0 simply represents the square of the sum of input data, and does not carry any epidemiological meaning regarding periodicity. Note that

the square root of the amplitude at 0, 10^{14}, is 10^7, which is of same order of magnitude as the total cases of COVID-19 in United States, which is about $4.3 \cdot 10^7$

3.2.2 Seven Days Period

Periodograms of COVID-19 daily new cases for 25 countries are shown in Figure 3.2. These graphs are the zoomed in periodograms, showing periods ranging from 1 day per cycle to 9 days per cycle, to emphasize the clear peak at 7 days per cycle.

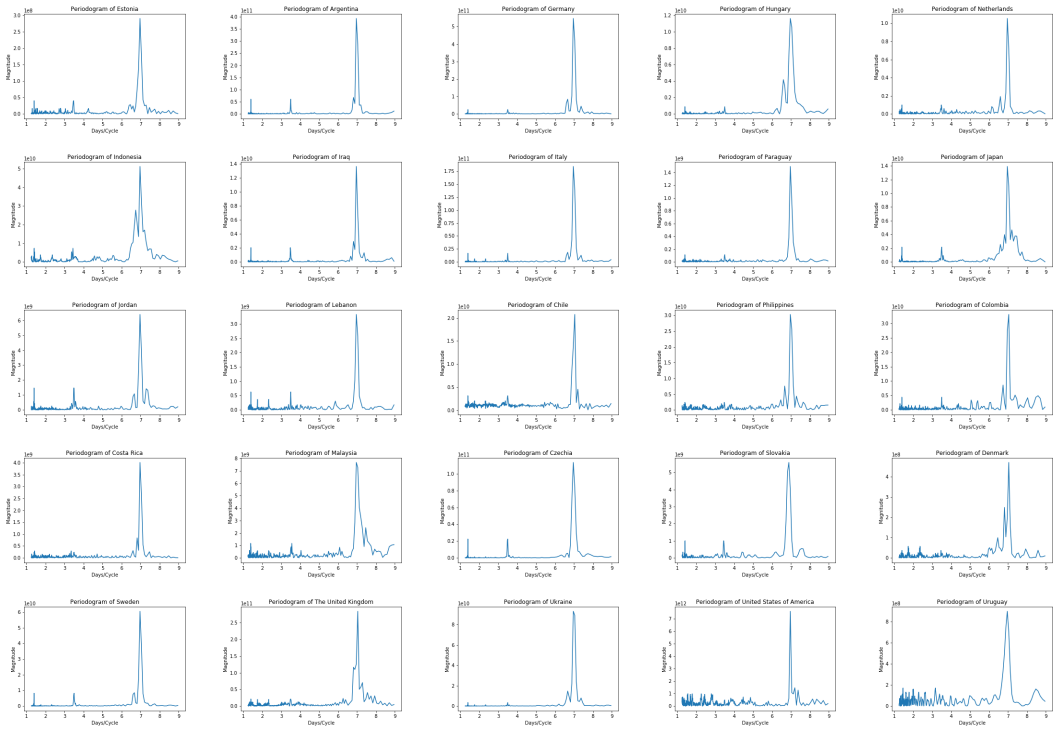

FIGURE 3.2: Zoomed In Periodogram for 25 Countries

The periodicities demonstrated in the periodograms are very close to one week. The less manifest periodicity of approximately 3.5 days per cycle can be attributed to its nature of being half of the length of a week. The weekly or half-weekly periodicity in daily new cases represents the weekly tempo of testing, as the oscillation of incidence data coincides with the oscillation of daily testing numbers (Bergman et al., 2020). Societal factors such as increased weekend activity are responsible for the 7 days periodicity, for example, increased testing on Wednesday clears the accumulated unconfirmed infectants from weekends, which manifests itself as a pseudo increase in reported cases on Wednesday. Such weekly periodic oscillation displays a shortcoming of this research, which is limited by the characteristics of the WHO Dataset. After all, human reporting errors or imperfect data collecting methodology are not factored into the analysis.

3.2.3 Twelve Months Period

Annual oscillation of daily new cases exists in COVID-19, as highlighted by the peaks around 300 days per cycle in Figure 3.3. The theoretical period at which the peak should occur is 365 days; the peaks in Figure 3.3 deviates from 365 days due to the lack of COVID-19 data to produce more precise results. The peaks on the periodograms actually occur when the frequency is 2, which translates to 313.5 days

per cycle, which is the data size, 627 days, divided by 2. However, this difference in annual cycle should not preclude the presence of seasonality in COVID-19, as a cycle length of 313.5 days is a close enough approximation of a year.

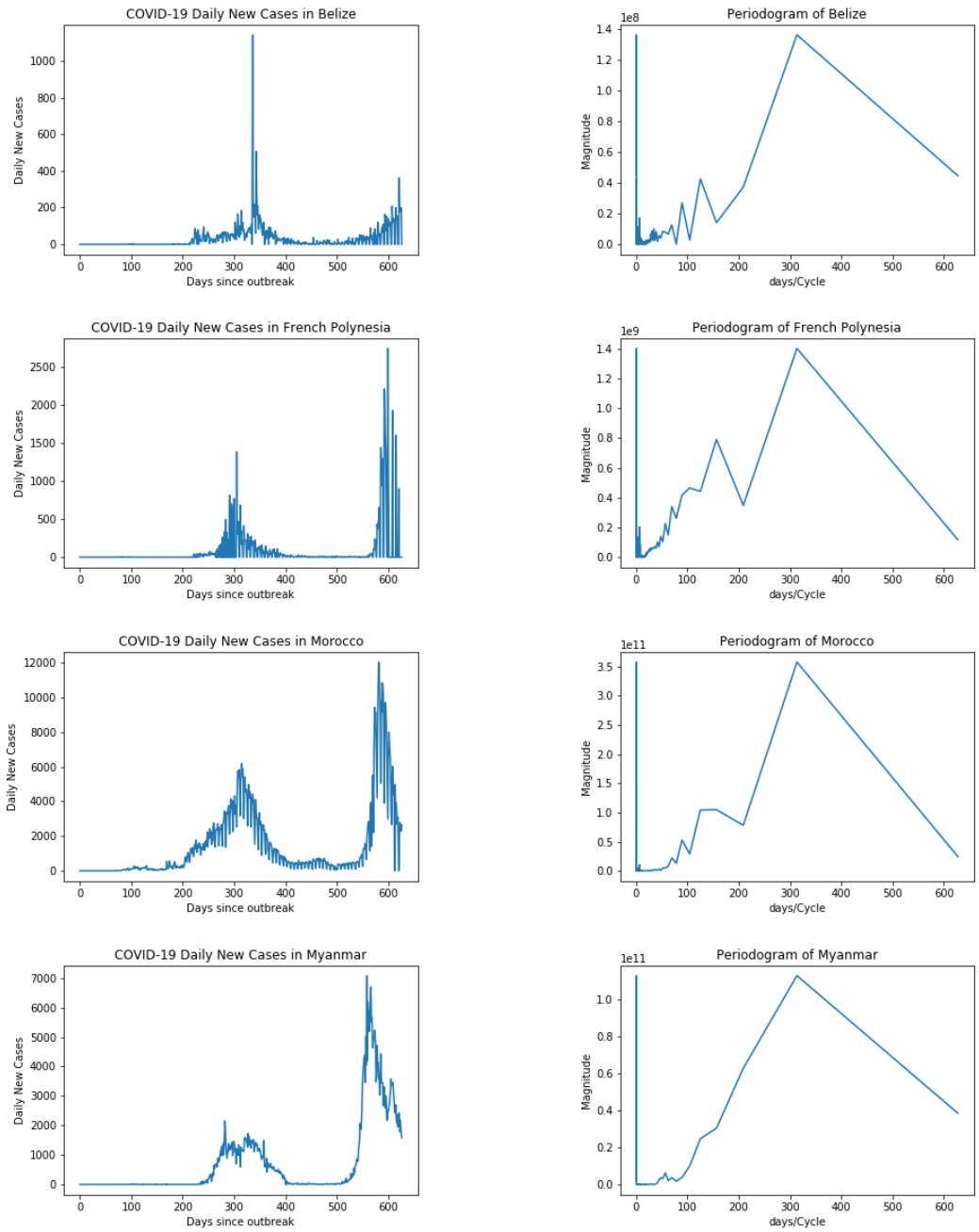

FIGURE 3.3: Time Series of COVID-19 Daily New Cases (left) and Their Corresponding Periodograms (right)

In the WHO Dataset, the first COVID-19 case was recorded on January 3, 2020; accordingly, the 300-days peak corresponds to the winter of 2020 and the 600-days peak corresponds to the winter of 2021. There are many explanations for the underlying mechanism of COVID-19's seasonality, but the different extents of their contributions remain debatable.

In colder weather, people spend more time indoors, potentially in a crowded public space with poor ventilation, hence boosting people's exposure to the virus(Fares,

2013). In winter, people are also more likely to have vitamin D deficiency, which is associated with increased susceptibility to infection(Reilly, 2020). Vitamin D is produced in the body through exposure to sunlight; however, with shorter sunlight hours and people wearing heavier clothing in winter, skin's exposure to sunlight is limited, leading to a lower Vitamin D concentration. The enhanced survival of viruses at low temperatures also contributes to the surge of infectants in winter by strengthening their infectiousness.

The effect of geographical location on seasonality is hard to identify, because intercountry interactions blur the influence of different climates on COVID-19. Although prohibiting oversea traffic and shutting down international borders can help contain the spread of disease to other countries, countries are not completely closed environments. Imported cases synchronize COVID-19 outbreaks in different countries, so the epidemic patterns in individual countries are altered and obscured.

3.2.4 One Frequency Peak

The periodograms and time series of daily new cases for 6 countries are shown in Figure 3.4. The distinct peak at 627 days per cycle, suggesting the entire data set forms one cycle, reveals the absence of a second outbreak of COVID-19 in certain countries.

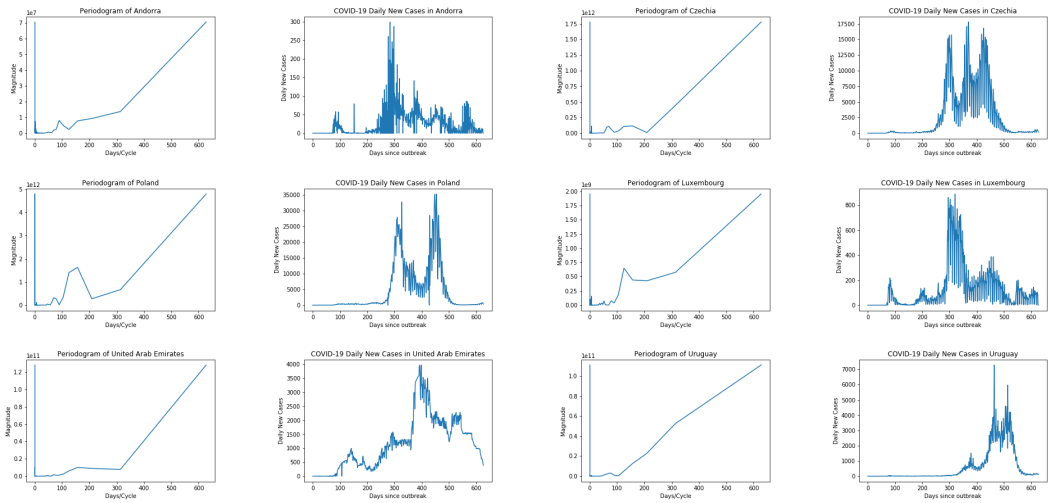

FIGURE 3.4: Time Series and Periodograms of COVID-19 New Cases

After the single peak in time series, daily new cases quickly decreased and have not changed significantly since then; a single peak in time series manifests as a peak at $frequency = 1$ in periodograms.

The lack of new cases after the first outbreak is a combination of the effect of high temperature in the summer and governments' success in trying to contain the spread of the virus through vaccination: generally, herd immunity was achieved in the summer of 2021 in these countries. The countries in the figure above are the leading countries in vaccination rates, all with more than 50% of the population vaccinated as of September 2021. Having only one peak in the periodogram testifies to the effectiveness of vaccines by indicating that there have been few infections after the first and only outbreak.

Vaccination has led to a change in the characteristics of COVID-19 in some countries: from a regular annual pandemic to an outbreak of only one surge. However,

as new COVID-19 variants appear and vaccines lose effectiveness, re-infections will occur, and the disease model will fall back under the influence of seasonality.

Chapter 4

Conclusion

4.1 Limitations in Research

Given that the COVID-19 pandemic has happended for less than 2 years, a key limitation to this research is the small data size available. The insufficiency of data casts uncertainty on the recurrent fluctuations in daily new cases, since more cycles will warrant more generality in pandemic behaviors. The limited dataset size also significantly reduces the accuracy and ability to detect periodicity. The difference between local infections and imported cases in daily new cases is not differentiated in this research; additionally, unreported occurrences of COVID-19 are not factored into the analysis.

4.2 Significance of Research

Frequency domain analysis provides solid proof of the periodicity of COVID-19, and several epidemic patterns of COVID-19 in different circumstances are discovered; more in-depth analysis of periodicity can potentially aid the government in interpolating future trends of COVID-19, impact the longterm planning of public health interventions, and prevent future COVID-19 outbreaks. The periodograms of COVID-19 daily new cases confirm that vaccination is powerful and beneficial in limiting the spread of COVID-19.

When the coronavirus undergoes unacknowledged mutations, the variation, which otherwise may be unrecognized, can be quickly spotted through the virus's deviation from its normal patterns in the periodogram.

4.3 Future Works

Future studies on the epidemic patterns of COVID-19 will benefit from a more abundant data set, which will make research results more accurate and may reveal additional periodicities. The periodograms of COVID-19 can be compared and contrasted with those of other epidemics to identify similarities and differences between their epidemic behavior. For example, frequency domain analysis can provide insights on whether COVID-19 will disappear after some years or continue to outbreak every year.

Appendix A

Resources

- Code Used in Research
- WHO COVID-19 Global Data
- LaTeX Template
- Interactive COVID-19 Data Base (Worldometer)

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